

GREEN BUILDING ON A TIGHT BUDGET

An innovative program in Austin, Texas, uses common sense ideas to build affordable green houses while providing job skills for disadvantaged youth.

By ELKA KARL

Many builders and homeowners have considered the benefits of going green, but they think green building is out of their budget range. While the benefits of green building—higher energy efficiency, better health for residents, lower impact on natural resources—are obvious, it's sometimes hard to keep the long-range benefits in mind when faced with the sticker shock of showboat green McMansions. However, if you're willing to think innovatively about your building process, and compromise on a smaller building envelope, you may be in for a pleasant surprise. Take a look, for instance, at some of the homes that have recently been built in East Austin, Texas. This low-income area of the music city capital of the world is dotted with affordable, attractive, and green homes that have been built over the past ten-odd years by a unique program called Casa Verde Builders. Program grants determined that the East Austin area would be the focus for home construction projects at Casa Verde.

A program of American YouthWorks (formerly the American Institute for Learning), Casa Verde Builders addresses several problems at once. This program provides disadvantaged youth with solid carpentry and other building skills, allows them to complete their high school diploma, and earn money for college through the national service organization AmeriCorps. They accomplish these goals while constructing affordable energy- and resource-efficient



Most of the youth who work with Casa Verde have had no prior building experience.

housing that is then sold to people who want to buy a new home but are working with a limited budget. The program focuses on at-risk youth aged 17–25 who are from low-income families and have an educational deficit; some of the students may have come from the judicial system. As of 2005, 1,204 youth have participated in the program. Of these, 765 were adjudicated, and 758 earned AmeriCorps awards.

Building It Green for Less

Chester Steinhauser, AmeriCorps programs coordinator at Casa Verde, started out in the organization as a math teacher

in American YouthWorks' General Educational Development (GED) program. He found that some of his students were having a difficult time conceptualizing the math problems they were studying, so he chose a practical approach to this difficulty: In 1993, he began taking his class out on Wednesday afternoons to a Habitat for Humanity house to use tape measures to get a better understanding for fractions. The practical approach worked so well that before long his students started telling him that they could build part of the Habitat for Humanity house. Around this same time, the newly formed Austin Green Building program was looking for a builder that could con-

Casa Verde's Best Features

Casa Verde-built homes are typically three-bedroom, two-bath, 1,200 ft² houses that contain a variety of energy efficiency and green building features.

Energy Efficiency Features

- 14-SEER central air conditioner
- Programmable thermostat
- Whole-house fan
- Ceiling fans in main rooms
- Continuous-ridge and soffit venting
- 2-ft overhangs sized to provide shade
- Cross ventilation
- Existing vegetation retained
- Galvalume metal roof
- Designed daylighting
- Large covered porch
- SIP exterior wall systems
- Double-paned windows w/solar screens

Recycled Content Materials

- Carpeting made of 100% recycled PET bottles
- Stramit compressed-straw panels
- Metal interior framing
- Galvalume metal roofing
- Old Paint primer (100% recycled paint)
- Choice Deck porch decking (100% recycled composite lumber)
- Cellulose insulation (100% recycled newsprint)
- High-fly-ash-content concrete

Indoor Environmental Quality Features

- Low-VOC paints and adhesives (less than 150 grams per liter)
- Medite formaldehyde-free MDF cabinets
- Pleated-media HVAC filters

Engineered Materials

- Manufactured lumber floor I joists
- Laminated veneer lumber rim joists and beams
- Manufactured roof trusses
- OSB Sturdi-Floor decking
- Hardie Company siding and trim products

struct an energy- and resource-efficient home, which was a requirement of one of its grants. After several months, the Green Building program partnered with Habitat for Humanity—and with American YouthWorks. Steinhauser's students now had the opportunity to walk their fraction talk. On this first project, the Green Building program provided technical support, Habitat for Humanity provided the lot, and American YouthWorks provided the building power. The house was built with the help of ten American YouthWorks members who were Steinhauser's students. They proved beyond a doubt that they could do fractions, and use those fractions to build a house.



Casa Verde Builders in Austin, Texas, builds affordable, green homes with the help of Austin youth.

Since that time, the program that became known as Casa Verde has built more than 70 houses. An average Casa Verde-built home is 1,200 ft², with three bedrooms and two bathrooms. These houses are designed to provide comfortable living space in a small home, with a focus on energy efficiency. Most homes feature front porches that provide shade during Austin's witheringly hot summers, and employ other cooling techniques, such as passive-solar design, high insulation levels, and a 16-SEER or better A/C unit.

"We look for the simple, easy things to do that don't cost us any money," notes Steinhauser. He adds that one of these cheap, easy fixes initially surprised the HVAC crew, who are professionals recommended by the Austin Green Building Program. All of the ductwork is placed in interior spaces.

"They thought we were crazy, but it's a really simple thing to do, and very cost-effective," says Steinhauser, noting that up to 30% of conditioning energy is lost in hot attics.

Other simple and cheap measures include installing fluorescent lamps in all homes, using finger-jointed studs, and placing studs every 24 inches, stacked directly beneath ceiling joists, instead of every 16 inches. This makes it possible to use a single top plate. Casa Verde also uses drywall clips, which eliminate the need for deadwood (blocking) on ceilings and walls.

Using drywall clips and other optimum-value engineering techniques saves Casa Verde up to 30% in framing costs.

Currently, Casa Verde Builders uses stick framing for its homes for economic reasons. Rising home values threatened to make its homes unavailable to its target low-income families, so a favorite but more expensive building component, structural insulated panels (SIPs), had to be replaced

to keep costs down. These high-performance building panels have expanded polystyrene cores sheathed on both sides with oriented strand board (OSB). Casa Verde prefers the Agri-Panel version, which has compressed wheat straw as the insulating core. But using these SIPs triples the cost of framing as compared to the cost of stick framing, using optimum-value engineering. Casa Verde hopes to use AgriPanel again in the future.

For the exterior walls, Casa Verde has also used autoclave-aerated concrete (AAC). Steinhauser has had great success with AAC at Casa Verde. It's an easy material for low-skilled construction crews to use because it cuts easily and accurately with a saw. AAC is a precast, manufactured building stone made of all-natural raw materials. It is an economical, environmentally friendly, and lightweight but structural material that

Hard Corps Learning

While Casa Verde's program affects the lives of first-time homeowners, it also affects another more specific population—its own workers. Stephanie Perrone, who is now a trainer for Casa Verde, began her career as an AmeriCorps member with Casa Verde in 2001. The next year she became a trainer, and she has been running job sites for the past three years. Perrone, who has a BS degree in architecture from the University of Nevada-Las Vegas, felt that working with Casa Verde was a logical choice for her education. "I wanted to have an opportunity to build a home and use sustainable products," she says. "This is a really nice way to get hands-on experience with green products."

Casa Verde member Nicole Garcia, who grew up in central Austin, is on Perrone's work crew. She has just started her second year of service with Casa Verde and recently finished her high school diploma at American YouthWorks' charter high school. "I had dropped out of school when I had my son," says Garcia. "I wanted to go back, and people I knew were going to the [American YouthWorks] school. I started school there in 2003. I had

applied to work with Casa Verde, and I signed up in January of last year."

Garcia is enthusiastic about her work and the effect it has had on her life. "I like it so much," she says. "Before I came out here, I was very shy and antisocial. Now I'm more open with people. I've also learned about home building. I never thought I could do this, and it makes me feel good to know that I can build homes."

Garcia knew nothing about green building or energy efficiency in homes before she signed on with Casa Verde. She says that the way Casa Verde builds houses, using techniques such as overhangs, passive solar design, pier-and-beam construction, and the use of green materials, has opened her eyes to the benefits of energy efficient and green building. From an economic point of view, she says, it just makes sense. "I compare my apartment bill with the bills of these homes," she says, "and I wish I would've known about this before."



TED ROAN

Through Casa Verde, Nicole Garcia has learned valuable construction skills while finishing her high school diploma and earning money for college.

One of Garcia's eventual goals is to become a homeowner herself. She will soon start school at Austin Community College, where she'll pursue a degree in nursing or another medical field.

When I ask Garcia what her five-year-old son thinks of his mom's line of work, she laughs. "He thinks it's crazy," she says. "It's kind of weird for him to see me come home all dirty and tired from working." But it seems obvious that he must admire his mom's work, too. "He's come out here to the job site to see what we do, and he likes it a lot out here. He likes to see what we do."

offers thermal and acoustic insulation. It is also fire and termite resistant. While AAC has been used throughout Europe for over 50 years, it has yet to be widely embraced by the U.S. building market.

Casa Verde also uses fiber cement board siding in its construction, most notably HardiPlank. Produced by the James Hardie Company, this siding is made using Portland cement, sand, and a very small amount of wood pulp. It is available in a variety of textures, including some that look like cedar, and it comes in different widths and lengths. It can be bought preprimed, unprimed, or already painted.

When Casa Verde member Jacob Johnson is asked about his favorite materials, he singles out the flooring system used in the houses. "I really like our floor system" he says enthusiastically,

"because of how efficiently wood is used. We build pier-and-beam houses, and for our beams we use laminated veneer lumber (LVL) beams that basically shave a tree by rotating it and then pressing all of these thin pieces together. For our joists we use I joists with LVL material on the top and bottom with an OSB sheet as the connecting piece. With LVL and the I joist you can order them to length, so nothing is wasted on site. The I joists are very light—two people can lift a 2 x 12 x 32—but just as strong, so not as many people are needed to hang them, which is another aspect of efficiency. Finally, our subfloor is 3/4-inch OSB, which is scrap wood and trash trees pressed together to make a sturdy floor. So we have a wood floor system that uses as little wood as possible."

Wheat straw panels are used for interior walls, and Casa Verde considers them one of its luxury items. For interior partitions, the 2 1/4-inch-thick panel can replace drywall and studs. Steinhäuser explains that Casa Verde likes to use wheat straw because it hopes to see this industry grow; by using wheat straw in affordable housing, it shows other builders that this is a worthwhile material. Using wheat straw also helps farmers, since it enables them to sell what for them is essentially a waste product. The only adhesives used in wheat straw panels are the natural resins in the straw that hold them together; boric acid is added to the straw as a fire retardant. Wheat straw panels have structural and tooling characteristics that are superior to those of the highest-grade wood-based particleboard, but they have several other advantages over particleboard. The raw

material is wheat straw—a waste product from fields that is available each year. Since the adhesive in wheat straw panels contains no formaldehyde, the indoor environmental quality (IEQ) of a home made with wheat straw panels is potentially much better than that of a home made with products containing formaldehyde. Finally, wheat straw panels are stronger, lighter, more moisture resistant, and easier to tool than particleboard.

Either cotton batt insulation, made from postconsumer jeans and treated with boric acid, or wet-spray cellulose insulation is used in walls and attics. Wet-spray cellulose insulation is used for stick frame walls. Casa Verde tries to maximize all site-delivered products and minimize what is sent to the landfill. This means preparing ahead of time, and using engineered and pre-cut materials so there is little waste wood. On projects, any extra wheat straw that is left over is scattered in the yard for mulch.

Case Verde and Austin Green Building

If you want to get the job done right, it's good to have a strong knowledge base to draw from. In the case of Casa Verde, this knowledge base comes in the form of the Austin Green Building program, the organization that originally teamed up with Casa Verde to build its first home. In the early 1990s, when Casa Verde and the Austin Green Building program were both forming, Katie Jensen was working as an AmeriCorps Vista member with American YouthWorks. While she was at YouthWorks, Jensen was able to use her position to gain a basic knowledge of affordable and green building. Today, she is the conservation program associate, and Leadership in Energy and Environmental Design (LEED) Accredited Professional (AP), at the Austin Green Building Program.

Casa Verde is about a lot more than real estate, says Jensen. "It's not just about the houses; it's also about the young people who are building their lives and rebuilding their community." Casa Verde has been very active in the Green Building program, and the reverse is also true. Austin Green Building's staff serve

as consultants for Casa Verde, and Casa Verde benefits from their recommendations for building materials, design, and other factors. Jensen praises Casa Verde's trainers and staff, who have used innovative design and materials while staying within their modest budget.

"Casa Verde's really neat because [the students] can see the fruits of their labor. They get to see the results of their work at the end of the day," says Jensen. The Austin Green Building program acts as a consultant for the public as a whole on energy efficiency and green building matters. It is the only program that



Casa Verde homes routinely receive at least three out of a possible five stars from the Austin Green Building Program's rating system for green homes.

requires mechanical calculation for green building using cooling equipment sized through a *Manual J* calculation. A copy of the *Manual J* printout, showing input and output values, must be attached to the Green Building rating for all projects. Casa Verde's homes are rated through the Green Building program on a scale of one through five stars. A one-star home has basic green building elements, while a five-star home is filled with a multitude of green elements. Homes are rated in five areas: energy efficiency, water efficiency, materials efficiency, health and safety, and community. Casa Verde also tracks energy use in homes after they are built, and its homes are air-tested as well. Casa Verde usually receives four or five stars on its homes.

An Asset to Austin

Each year Casa Verde reapplies for YouthWorks and AmeriCorps awards, while also receiving support from other

organizations and corporations such as Home Depot. The Austin Green Building program also provides Casa Verde with technical assistance for its homes. The group has been nationally recognized for its work. Awards include the Points of Light Foundation Presidential Service Award; the Promising and Effective Practices Network (PEP-Net) Effective Initiative Award; the Texas Society of Architects Citation of Honor; the U.S. Department of Commerce HUD Best Practices Award; and the National Association of Home Builders (NAHB) Affordable Category Hot Climate Gold Award (2005).

The community as a whole has benefited from the homes as well. Each year, the city of Austin collects over \$150,000 in property taxes from the houses built by Casa Verde. Indirectly, the city also benefits from the increased energy efficiency of these homes, which lowers overall energy use, especially in the hot summer months. Steinhäuser also notes that this area of East Austin was drug and crime ridden only ten years ago. Now, thanks in no small part to the construction of the Casa Verde homes, working couples and families are moving into the neighborhood, making it more stable and a more pleasant place to live.

Elka Karl is an associate editor with Home Energy magazine.

FOR MORE INFORMATION:

For information on Casa Verde Builders' innovative green building program, or for information on joining its YouthWorks AmeriCorps program, contact Chester Steinhäuser at (512)744-1940 or e-mail him at csteinhauser@americanyouthworks.org. You can also read about the program at www.aill.org/cvb.html.

For information about Austin Green Building, call (512)482-5300, or go to www.ci.austin.tx.us/greenbuilder/.